

JOSEPH HENRY

US scientist and engineer Joseph Henry was at the forefront of advances in electromagnetic technologies and helped establish the modern electrical industry.



From humble beginnings, Henry (1797–1878) became a leading scientist due to his work on electromagnetism. Independently of Faraday, he discovered that a moving magnetic field can “induce” an electric current. He built the most powerful electromagnets in the world, pioneered the technology that led to the electric telegraph, and was technical advisor to Abraham Lincoln. The First Secretary of the Smithsonian Institution, Henry’s work also led to the creation of the US National Weather Service.

benzene), and became the first to liquify a permanent gas (one that was believed to be impossible to liquify). Yet it was the field of physics that captured Faraday’s imagination and that inspired his key contributions to science.

Discoveries in electromagnetism

In 1820, the Danish scientist Hans Ørsted found a link between electricity and magnetism—namely that when a current is passed through a wire, it produces a magnetic field around the wire. This prompted Faraday’s first great discovery. By suspending an electric wire into a cup of mercury with a magnet at the bottom, Faraday demonstrated that if a current

was run through the wire, the magnet would move around it in a circle. His results, published in 1821, proposed that electromagnetic energy could produce continuous motion: Faraday had created the world’s first electric motor.

A decade later, Faraday made an even more significant discovery—that moving a magnetic field can cause electricity to flow through a conductor. In a groundbreaking experiment in which a copper disk was spun between the two poles of a magnet, Faraday generated a steady electric current and produced the first dynamo, or electric generator. This invention had huge implications: during the early 19th century, electricity was the sole authority of scientists and had no real practical application. But Faraday had discovered a means of generating electricity without a battery, and in vast quantities. His method remains the basis on which modern power plants operate.

Setting the modern stage

Faraday was a visionary who proved his theories via experimentation, such as the 1845 experiment in which he discovered the effects of magnetism on light. However, some of his ideas proved too radical, including his theory of the “unity of forces,” which stated that magnetism, light, electricity, and gravity

TOOK 10 YEARS TO SUCCESSFULLY CONNECT MAGNETISM TO ELECTRICITY IN 1831



SPENT 54 YEARS WORKING AT THE ROYAL INSTITUTION

LAID THE BASIS FOR FUTURE DEVELOPMENTS IN THE FIELD OF ELECTROMAGNETISM

